

See surplus. Detect need. Coordinate impact.

Leveraging geospatial intelligence
for equitable medical-supply redistribution

Lesnar Gitonga · USIU-Africa · 23 June 2026

Conference demo · synthetic inventory · no patient records

The problem is not only scarcity. It is disconnected visibility.

Facilities can hold usable surplus while others face urgent shortfalls. MediMatch makes both sides visible in one operating picture.

56.3%

face weekly or monthly stockouts

82.8%

rate manual methods ineffective or slow

57.8%

report wastage of valid supplies

89.1%

willing to pilot the system

A supply gap becomes a coordination question

Which available source is close enough, sufficiently stocked, verified, product-compatible—and practical to move now?

57%

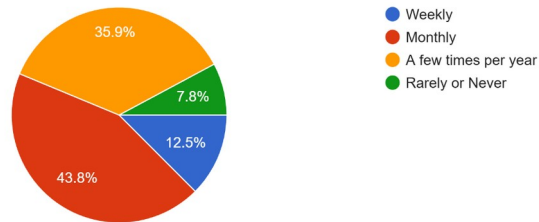
KEMSA national order-fill rate,
as of mid-2025 (per the abstract)

The baseline is historical; the deck does not present it as the current 2026 rate.

Survey results from 64 healthcare professionals

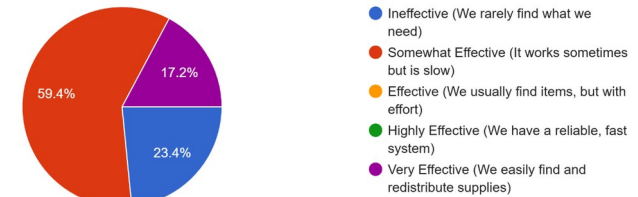
Nairobi County field study · 80% response rate · unedited Google Forms output

Q4. How often does your facility experience stockouts of essential items?
64 responses



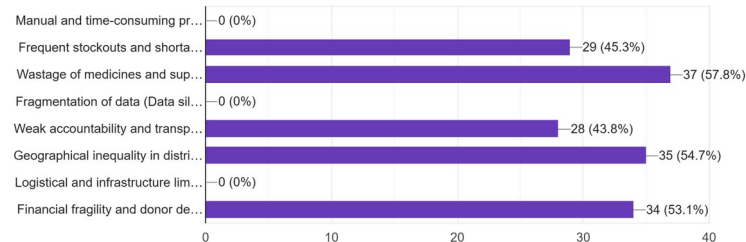
56.3% face weekly or monthly stockouts

Q6. How would you rate the effectiveness of your current method for locating emergency supplies from nearby facilities?
64 responses



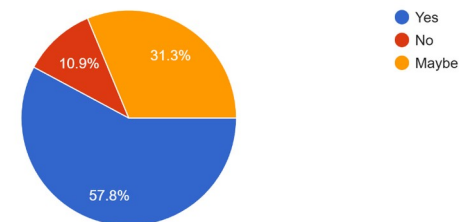
82.8% rate current methods ineffective or slow

Q7. Which of the following do you consider the main challenges of the current supply system?
64 responses



57.8% cite wastage of valid supplies

Q9. Would you recommend your facility explore piloting the MediMatch platform?
64 responses

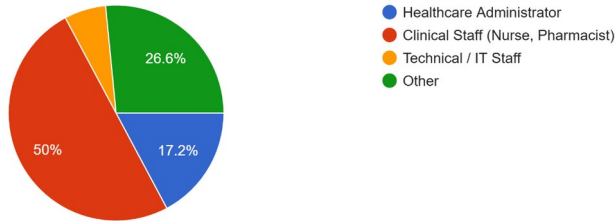


89.1% would pilot the system (Yes + Maybe)

Who we asked — and how facilities cope today

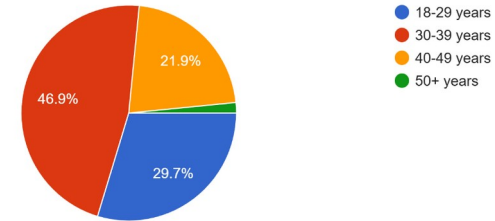
Nairobi County field study · 64 respondents · unedited Google Forms output

Q3. What is your primary role at this facility?
64 responses



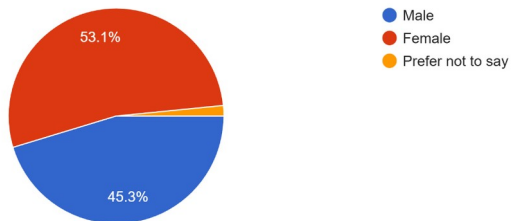
Mostly clinical staff — nurses & pharmacists

Q2. What is your age bracket?
64 responses



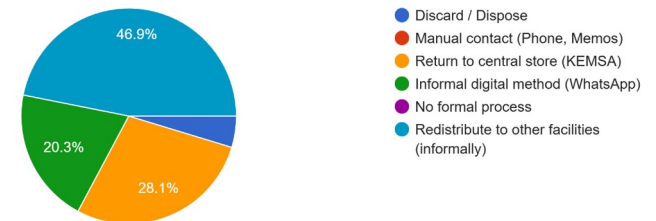
A predominantly under-40, digitally-fluent workforce

Q1. What is your gender?
64 responses



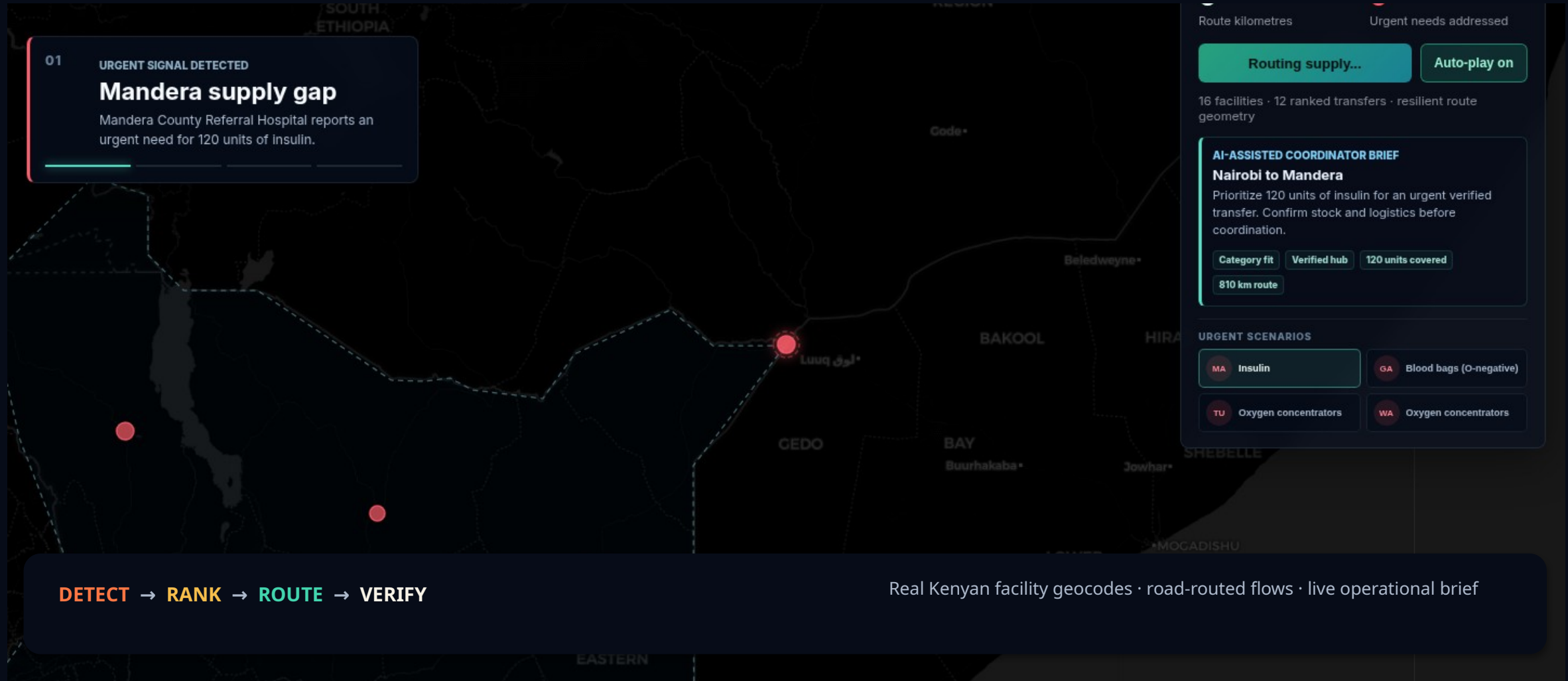
Balanced gender, slight female majority

Q5. What is your facility's current method for handling surplus (near-expiry) supplies?
64 responses



Today: ad-hoc — informal redistribution, KEMSA returns, WhatsApp

From isolated reports to one national operating picture

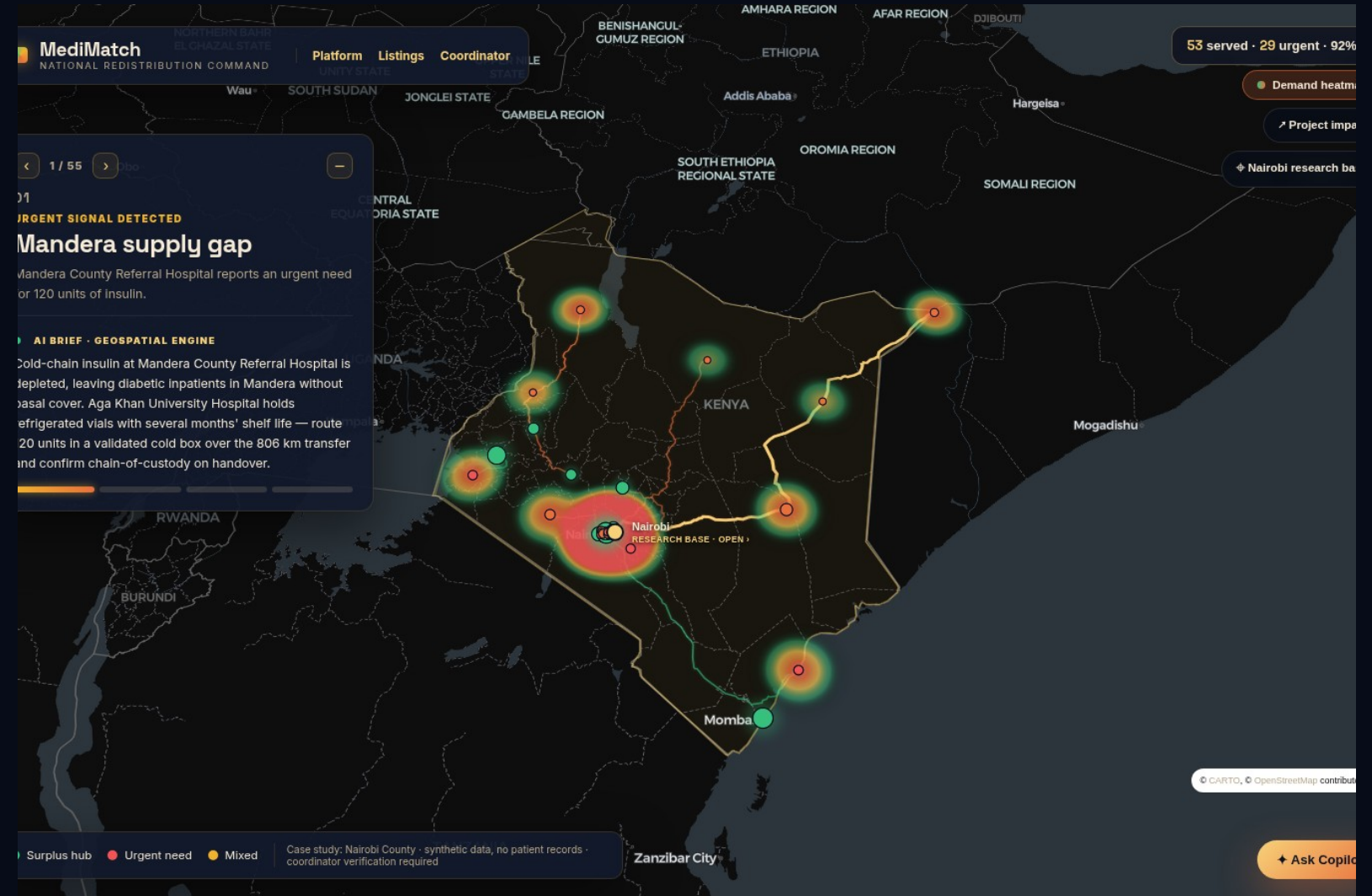


See demand concentration before routing

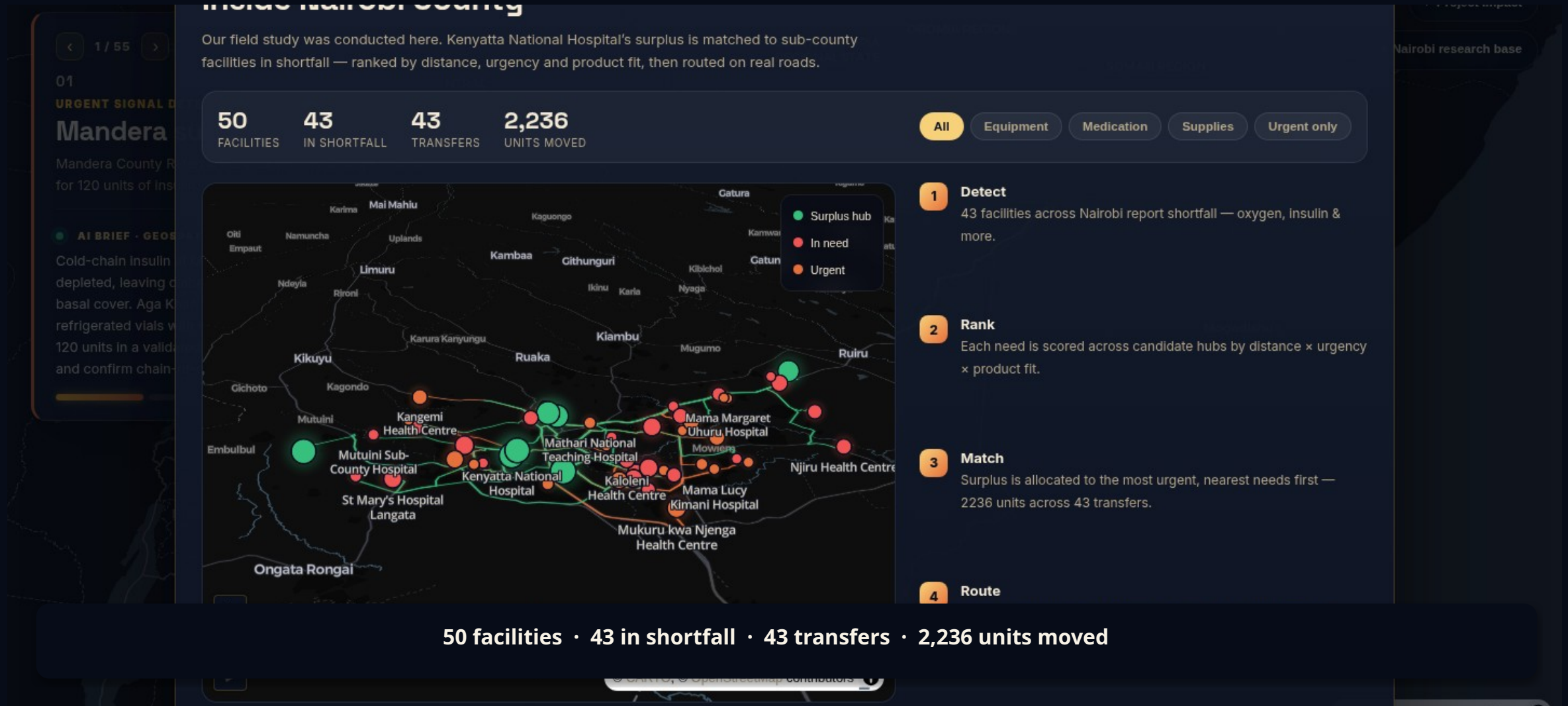
What geography adds

- Reveal where shortfall clusters—not just where the loudest request originated.
- Compare demand against available hubs and connected road corridors.
- Use distance as one signal alongside urgency, product fit and verification.
- Move from 'nearest' to 'best feasible match'.

Geography supports triage; it does not authorize a transfer.



Nairobi: the national model at street scale

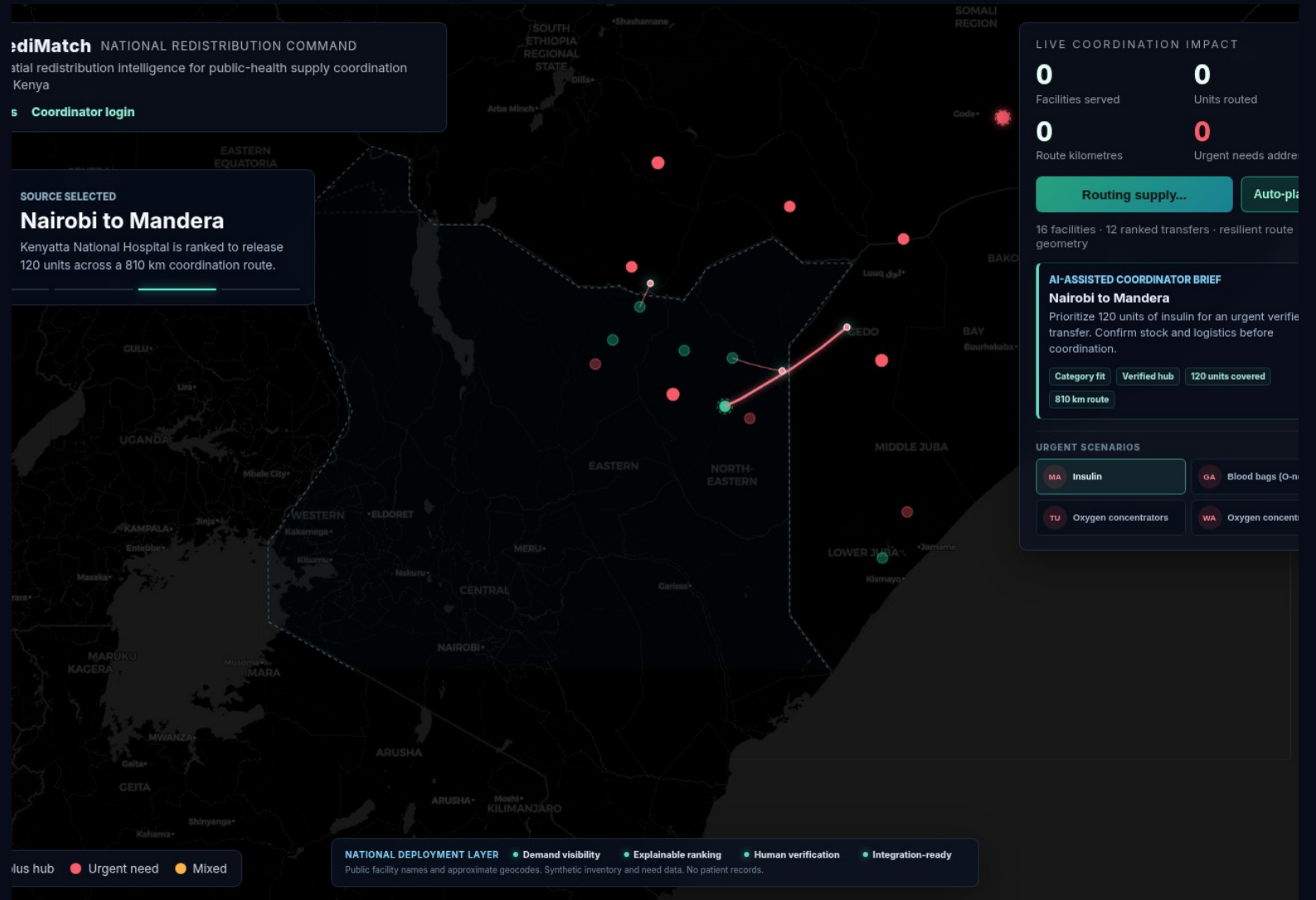


The system shows why a source was selected

Decision signals

- Proximity** direct and routed distance
- Urgency** clinical supply priority
- Product fit** category and cold-chain needs
- Quantity** full or partial coverage
- Verification** trusted source and ownership

Curated situation briefs explain the route in plain operational language.



What the current throughput could unlock over one year



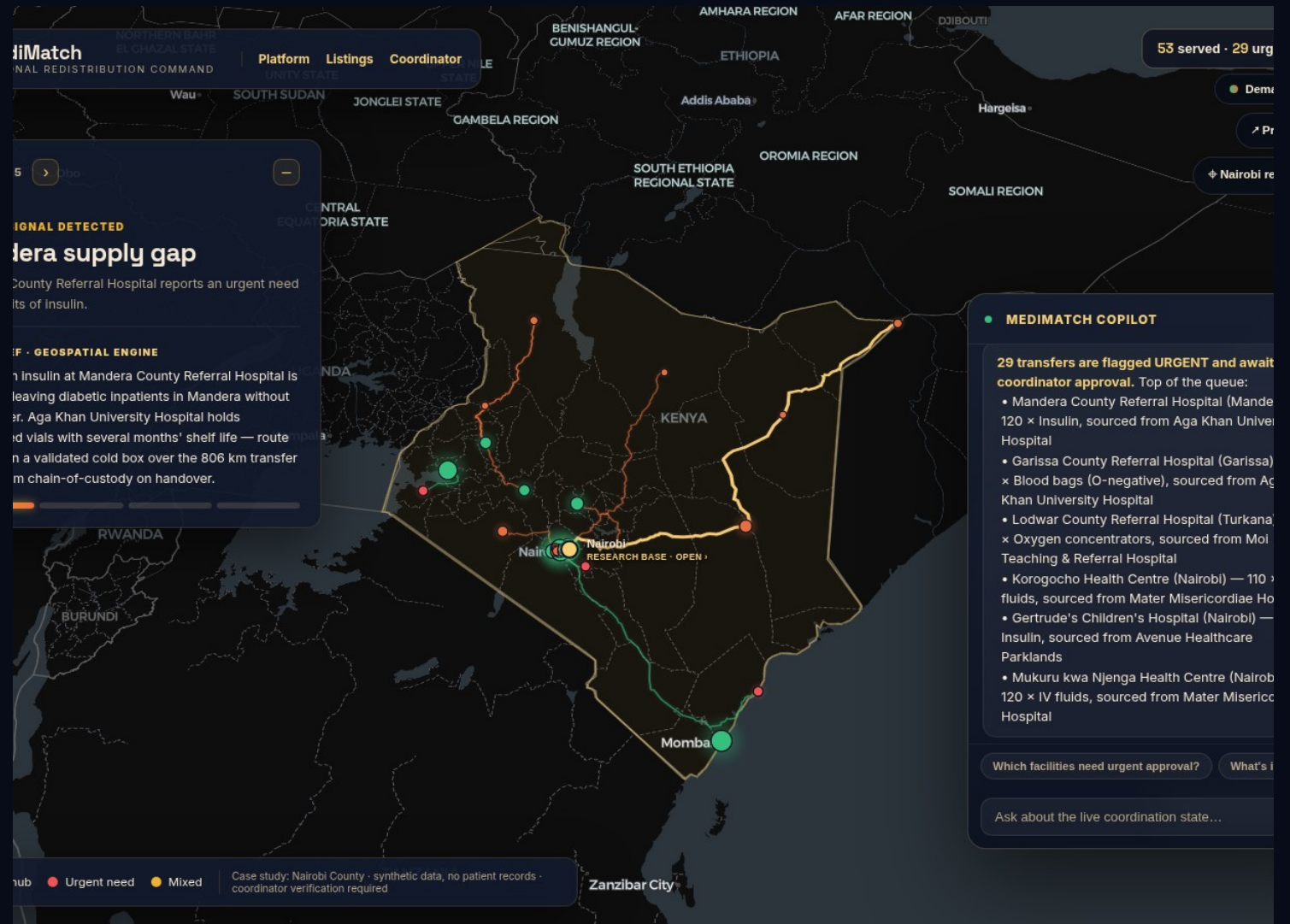
MODEL — NOT A GUARANTEE. Weekly cycles, 42% near-expiry recovery, coverage saturates as facilities onboard.

Coordination intelligence, grounded in live state

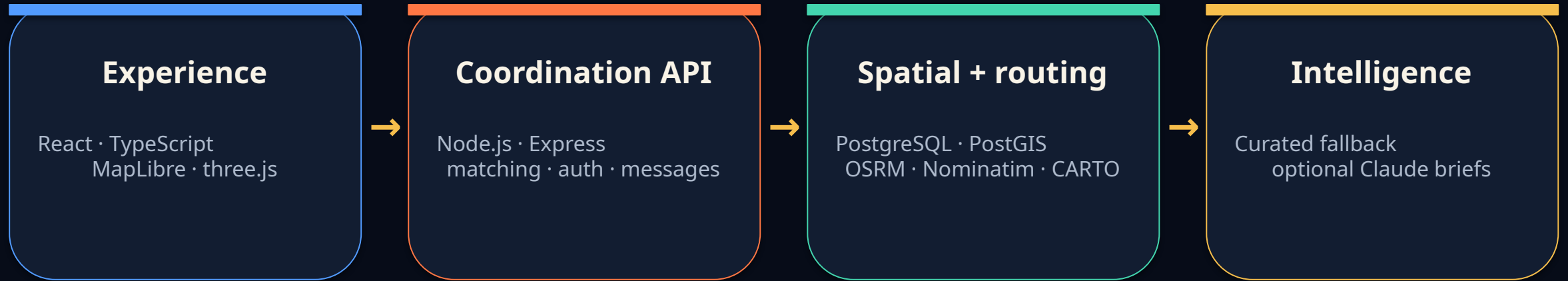
What the Copilot surfaces

- Facilities awaiting urgent approval
- Transfers currently in transit
- Shortfalls located by county
- Hubs holding redistributable surplus
- A concise national situation summary

Answers are drawn from the live coordination state, not a general model. The demonstration runs without an API key; an Anthropic key enables Claude-written briefs.



A production-minded spatial stack with a human decision boundary



Human verification is the final control

No diagnosis · no autonomous transfers · synthetic demo inventory · explicit assumptions · coordinator sign-off

Location is not just a field. It is a coordination advantage.

Evidence-based

64 healthcare professionals quantified the coordination gap.

Location-aware

Real-time, road-routed matching of surplus to urgent need.

Explainable

Transparent, multi-factor scoring — no black box.

Human-governed

Coordinator verification before every transfer.

MEDIMATCH

**See surplus. Detect need.
Coordinate impact.**

Thank you